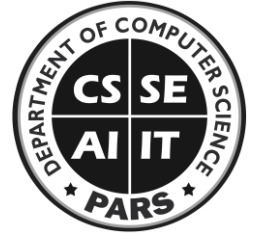




UNIVERSITY OF AGRICULTURE

Department of Computer Science



Lab 4: Programming Fundamentals

(CS-308)

Basic Information			
Lab #:	4	Teacher:	Arsal Mahmood
Objectives	1. While Loop 2. Do-while Loop		
Registration#	2025-ag-9987	Student Name	Syed Rameez Haider Rizvi
Total Marks	10	Marks Obtained	
Tools	Dev C++ / Visual Studio		

Activity 1 : While Loop

1. Calculate the sum of numbers from 1 to 100 and display it on screen.

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int i = 1, sum = 0;
```

```
    while (i <= 100) {
```

```
        sum += i;
```

```
        i++;
```

```
    }
```

```
    cout << sum << endl;
```

```
    return 0;
```

```
}
```

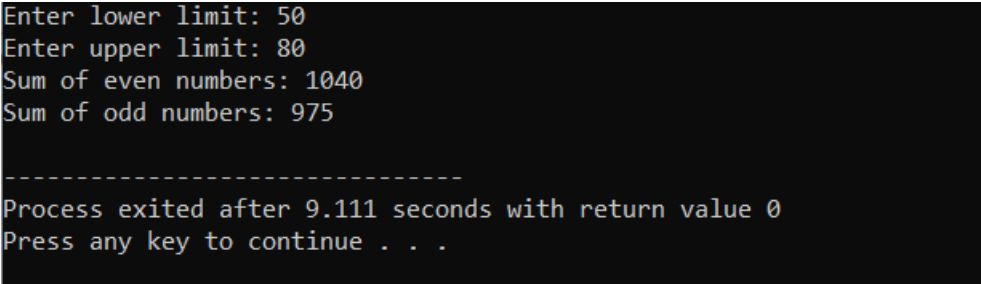
```
5050
```

```
-----  
Process exited after 1.27 seconds with return value 0  
Press any key to continue . . .
```

2. Calculate sum of even and off numbers separately from a user given range. Like if user gives **lower_limit** of 50 and **higher_limit** of 80. Calculate sum of all odds and evens in between 50 and 80 and display on screen.

```
#include <iostream>
using namespace std;

int main() {
    int lower, upper;
    cout << "Enter lower limit: ";
    cin >> lower;
    cout << "Enter upper limit: ";
    cin >> upper;
    int sumEven = 0, sumOdd = 0;
    int i = lower;
    while (i <= upper) {
        if (i % 2 == 0)
            sumEven += i;
        else
            sumOdd += i;
        i++;
    }
    cout << "Sum of even numbers: " << sumEven << endl;
    cout << "Sum of odd numbers: " << sumOdd << endl;
    return 0;
}
```



```
Enter lower limit: 50
Enter upper limit: 80
Sum of even numbers: 1040
Sum of odd numbers: 975

-----
Process exited after 9.111 seconds with return value 0
Press any key to continue . . .
```

3. Input the marks of a class of ten students. The marks must be between **0** to **100**. Also check marks should not be less than 0 or greater than 100. Calculate the grades of every student and display it. Also calculate the total of the grades and the class average.
- $>90 = A$

- <90 and $>80 = B$
- <80 and $>70 = C$
- <70 and $>60 = D$
- $<60 = F$

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
    int marks, total = 0, i = 1;
    char grade;
    while (i <= 10) {
        cout << "Enter marks for student " << i << ": ";
        cin >> marks;
        if (marks < 0 || marks > 100) {
            cout << "Invalid marks. Enter a value between 0 and 100." << endl;
            continue;
        }
        total += marks;
        if (marks > 90)
            grade = 'A';
        else if (marks > 80)
            grade = 'B';
        else if (marks > 70)
            grade = 'C';
        else if (marks > 60)
            grade = 'D';
        else
            grade = 'F';
        cout << "Grade of student " << i << ": " << grade << endl;
        i++;
    }
    cout << "Total marks: " << total << endl;
    cout << "Class average: " << total / 10.0 << endl;
    return 0;
}
```

```

Enter marks for student 1: 90
Grade of student 1: B
Enter marks for student 2: 95
Grade of student 2: A
Enter marks for student 3: 79
Grade of student 3: C
Enter marks for student 4: 07
Grade of student 4: F
Enter marks for student 5: 82
Grade of student 5: B
Enter marks for student 6: 56
Grade of student 6: F
Enter marks for student 7: 67
Grade of student 7: D
Enter marks for student 8: 78
Grade of student 8: C
Enter marks for student 9: 78
Grade of student 9: C
Enter marks for student 10: 78
Grade of student 10: C
Total marks: 710
Class average: 71

```

```

-----
Process exited after 77.94 seconds with return value 0
Press any key to continue . . .

```

4. Input a number from user and display a pattern like these on screen. For example, user has entered 5, the program should display following pattern on screen.:

(a)

```

    *      *      *      *      *
  *      * *      *
 *      * *
*      *
*
```

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int n, i, j;
```

```
    cout << "Enter a number: ";
```

```
    cin >> n;
```

```
    i = n;
```

```
    while (i >= 1) {
```

```
        j = 1;
```

```
        while (j <= i) {
```

```
            cout << "*" << "t";
```

```
            j++;
```

```

    }
    cout << endl;
    i--;
}
return 0;
}

```

```

Enter a number: 5
*   *   *   *   *
*   *   *   *   *
*   *   *   *   *
*   *   *   *   *
*   *   *   *   *
*
-----
Process exited after 10.06 seconds with return value 0
Press any key to continue . . .

```

P-T-O

(b) 1 2 3 4 5

1 2 3 4

1 2 3

1 2

1

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int n, i, j;
```

```
    cout << "Enter a number: ";
```

```
    cin >> n;
```

```
    i = n;
```

```

while (i >= 1) {

    j = 1;

    while (j <= i) {

        cout << j << "\t";

        j++;

    }

    cout << endl;

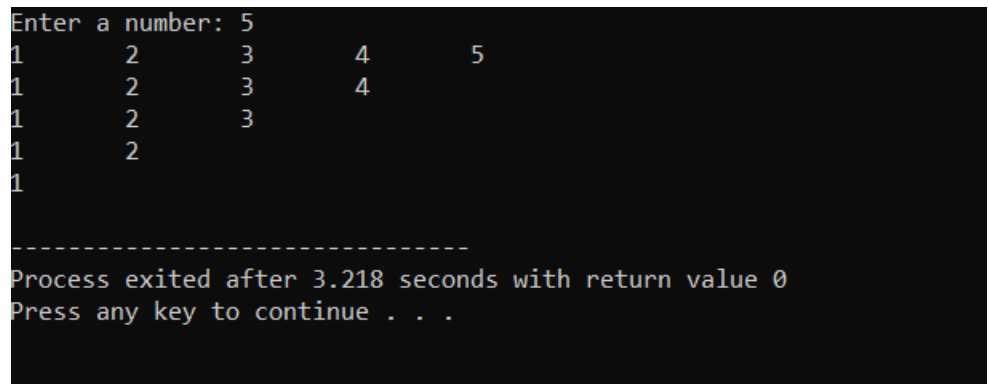
    i--;

}

return 0;

}

```



```

Enter a number: 5
1      2      3      4      5
1      2      3      4
1      2      3
1      2
1
-----
Process exited after 3.218 seconds with return value 0
Press any key to continue . . .

```

(c) 1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

```
#include <iostream>
```

```
using namespace std;
```

```

int main() {

    int n, i = 1, j;

    cout << "Enter a number: ";

    cin >> n;

    while (i <= n) {

        j = 1;

        while (j <= i) {

            cout << j << "\t";

            j++;

        }

        cout << endl;

        i++;

    }

    return 0;

}

```

The screenshot shows the output of the C++ program. It prompts the user to enter a number, and the user enters 5. The program then prints a pattern of numbers where each row i contains the numbers 1 through i, separated by tabs. The output is as follows:

```

Enter a number: 5
1
1      2
1      2      3
1      2      3      4
1      2      3      4      5

```

Below the pattern, a dashed line separates it from the program's exit message:

```

-----
Process exited after 3.764 seconds with return value 0
Press any key to continue . . .

```

(d) ? ? ? ? ?

 ? ? ? ? ?

?	?	?	?	?
?	?	?	?	?
?	?	?	?	?

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int n, i = 1, j;
```

```
    cout << "Enter a number: ";
```

```
    cin >> n;
```

```
    while (i <= n) {
```

```
        j = 1;
```

```
        while (j <= n) {
```

```
            cout << "?\t";
```

```
            j++;
```

```
        }
```

```
        cout << endl;
```

```
        i++;
```

```
    }
```

```
    return 0;
```

```
}
```



```
Enter a number: 5
?      ?      ?      ?      ?
?      ?      ?      ?      ?
?      ?      ?      ?      ?
?      ?      ?      ?      ?
?      ?      ?      ?      ?

-----
Process exited after 3.295 seconds with return value 0
Press any key to continue . . .
```

(Solve all these questions separately. Also, the pattern would change if the user given number is changed)

Activity 2 : Do-While loop

5. Write a program which inputs from user a character between **a** to **z** and guess whether the character matches with a pre-defined letter or not. If it matches, then display a message "**Congratulations**". It gives the user five chances or tries to guess the character.

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    char guess;
```

```
    char secret = 'k';
```

```
    int attempts = 1;
```

```
do {

    cout << "Enter a character between a to z: ";

    cin >> guess;

    if (guess == secret) {

        cout << "Congratulations" << endl;

        break;

    } else {

        cout << "Try again" << endl;

    }

    attempts++;

} while (attempts <= 5);

if (attempts > 5) {

    cout << "Sorry! You have used all attempts." << endl;
```

```

    }

    return 0;

}

```

```

Enter a character between a to z: k
Congratulations

-----
Process exited after 3.527 seconds with return value 0
Press any key to continue . . .

```

2

6. Calculate sum of even and off numbers by using do-while loop separately from a user given range. Like if user gives **lower_limit** of 50 and **higher_limit** of 80. Calculate sum of all odds and even in between 50 and 80 and display on screen.

```

#include <iostream>
using namespace std;

int main() {
    int lower, upper;
    cout << "Enter lower limit: ";
    cin >> lower;
    cout << "Enter upper limit: ";
    cin >> upper;
    int sumEven = 0, sumOdd = 0;
    int i = lower;
    do {
        if (i % 2 == 0)
            sumEven += i;
        else

```

```

        sumOdd += i;
        i++;
    } while (i <= upper);
    cout << "Sum of even numbers: " << sumEven << endl;
    cout << "Sum of odd numbers: " << sumOdd << endl;
    return 0;
}

```

```

Enter lower limit: 50 80
Enter upper limit: Sum of even numbers: 1040
Sum of odd numbers: 975

-----
Process exited after 5.711 seconds with return value 0
Press any key to continue . . .

```

7. Print the table of user given number from 1 upto user given limit. Like if user enters 3, and limit is 4, program should print table of 3 as follows:

3	*	1	=	03
3	*	2	=	06
3	*	3	=	09
3	*	4	=	12

```
#include <iostream>
```

```
using namespace std;
```

```

int main() {
    int num, limit, i = 1;
    cout << "Enter a number: ";
    cin >> num;
    cout << "Enter the limit: ";
    cin >> limit;
    do {
        cout << num << "\t*\t" << i << "\t=\t" << num * i << endl;
        i++;
    } while (i <= limit);
    return 0;
}

```

```
}
```

```
Enter a number: 3
Enter the limit: 4
3      *      1      =      3
3      *      2      =      6
3      *      3      =      9
3      *      4      =     12

-----
Process exited after 3.271 seconds with return value 0
Press any key to continue . . .
```

8. Ask the user to input numbers repeatedly until they enter -1.

- Keep track of the largest number entered.
- Print the largest number at the end.

Explanation of code:

while loop continues until -1 is entered.

Inside loop, use if condition to compare each number with current maximum.

```
#include <iostream>

using namespace std;

int main() {
    int num, largest;
    cout << "Enter a number (-1 to stop): ";
    cin >> num;
    largest = num;
    if (num == -1) {
        cout << "No numbers were entered." << endl;
        return 0;
    }
    do {
        if (num > largest)
            largest = num;
        cout << "Enter a number (-1 to stop): ";
        cin >> num;
    } while (num != -1);
    cout << "Largest number entered: " << largest << endl;
```

```
return 0;
```

```
}
```

```
Enter a number (-1 to stop): 5
Enter a number (-1 to stop): 12
Enter a number (-1 to stop): 8
Enter a number (-1 to stop): -1
Largest number entered: 12

-----
Process exited after 7.865 seconds with return value 0
Press any key to continue . . .
```

{end of Lab}